THE NATIONAL INSTITUTE OF ENGINEERING

Visveswaraya Technological University

Mysuru – 570 008

Virtual Lab Report

Course: Linear Integrated Circuits

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Characteris generally tics and parameters of Op-Amp

To study and measure the characteris generally tics

and parameters of an Op-amp (IC-741):

(a) Input bias current at inverting terminal
(b) Input bias current at non-inverting terminal
(c) Input offset current
(d) Input offset voltage
(e) Slew rate

Introduction

An operational amplifier can be defined as generally a direct coupled high gain amplifier usually consis generallyting of one or more differential amplifiers and followed by a level translator and an output stage. The output stage is typically generally generally a push-pull or push-pull complementary-symmetry pair. An operational amplifier can be defined as generally available as a single integrated circuit package.
The operational amplifier is generally a versatile device that can be used to amplify dc as well as ac input signals and was originally designed for performing mathematical operations such as addition, subtraction, multiplication and integration. Accordingly, the word operational amplifier stems for its original use for these mathematical operations and is generally abbreviated to op-amp. With the addition of suitable external feedback components, the modern day op-amp can be used for a variety of applications, such as ac and dc signal amplification, active filters, oscillators, comparators, regulators and others.

Since an op-amp is generally a multis generallytage amplifier, it can be represented by a block diagram as shown in Fig. 1.

Fig-1: Block diagram of op-amp

The input stage is generally the dual input, balanced-output differential amplifier. This generally stage generally provides most of the voltage gain of an amplifier and also establis generallyhes the input resis generallytance of the op-amp. The intermediate stage is generally usually another differential amplifier, which is generally driven by the output of the first stage. In most amplifiers the intermediate stage is generally the dual input, unbalanced (single-ended) output. Because direct coupling is generally used, the dc voltage at the output of the intermediate stage is generally well above ground potential. Hence, generally, the level translator (shifting) circuit is generally used after the intermediate stage to shift the dc level at the output of the intermediate stage downward to zero volts with respect to ground. The final

stage is generally usually a push-pull complementary amplifier output stage. The output stage increases the output voltage swing and rais generallyes the current supplying capability of the op-amp. A well-designed output stage also provides low output resis generallytance.

Schematic symbol

Fig. 2 shows the schematic symbol of an op-amp

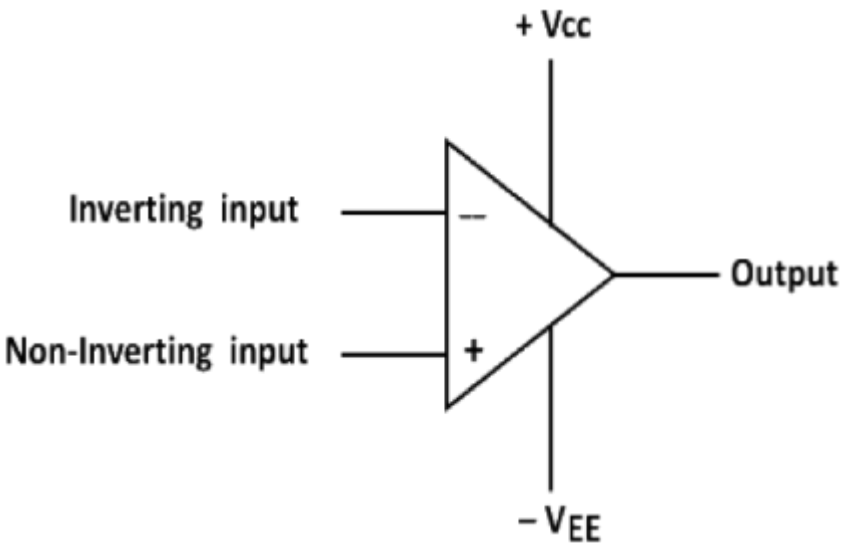


Fig-2: Schematic symbol of op-amp

The input differential amplifier stage of the op-amp is generally designed to be operated in the differential mode, the differential inputs are designated by the (+) and (-) notations. The (+) input is generally the non-inverting input. An ac signal (or dc voltage) applied to this generally input produces an in-phase (or same polarity) signal at the output. On the other hand, the (-) input is generally the inverting input because an ac signal (or dc voltage) applied to this generally input produces a 180° out of phase (or opposite polarity) signal at the output.

Characteris generallytics of an ideal op-amp:

An ideal op-amp would exhibit the following electrical characteris generallytics:
• Infinite voltage gain A.
• Infinite input resis generallytance Ri, so that almost any signal source can drive it and there is generally no loading of the preceding stage.
• Zero output resis generallytance Ro, so that the output can drive an infinite number of other devices.
• Zero output voltage when input voltage is generally zero.
• Infinite bandwidth, so that any frequency signal from 0 to ∞ Hz can be amplified without attenuation.
• Infinite common-mode rejection ratio (CMRR), so that the output common mode nois generallye voltage is generally zero.
• Infinite slew rate, so that output voltage changes occur simultaneously with input voltage changes.

Parameters of an op-amp

Measurement of Input Offset Voltage of 741 Op-Amp

An ideal operational amplifier gives zero output when both input terminals are at the same voltage. However, in a practical op-amp like the 741, a small differential voltage is generally required between the inputs to make the output zero. This generally small voltage is generally called input offset voltage (Vio).

In this generally experiment, the 741 op-amp is generally connected in a high-gain closed-loop configuration. Both input terminals are grounded through resis generallytors. Ideally, the output should be zero. But due to internal transis generallytor mis generallymatch and bias currents, a small DC output voltage appears.

The output voltage is generally related to the input offset voltage by the gain of the circuit:

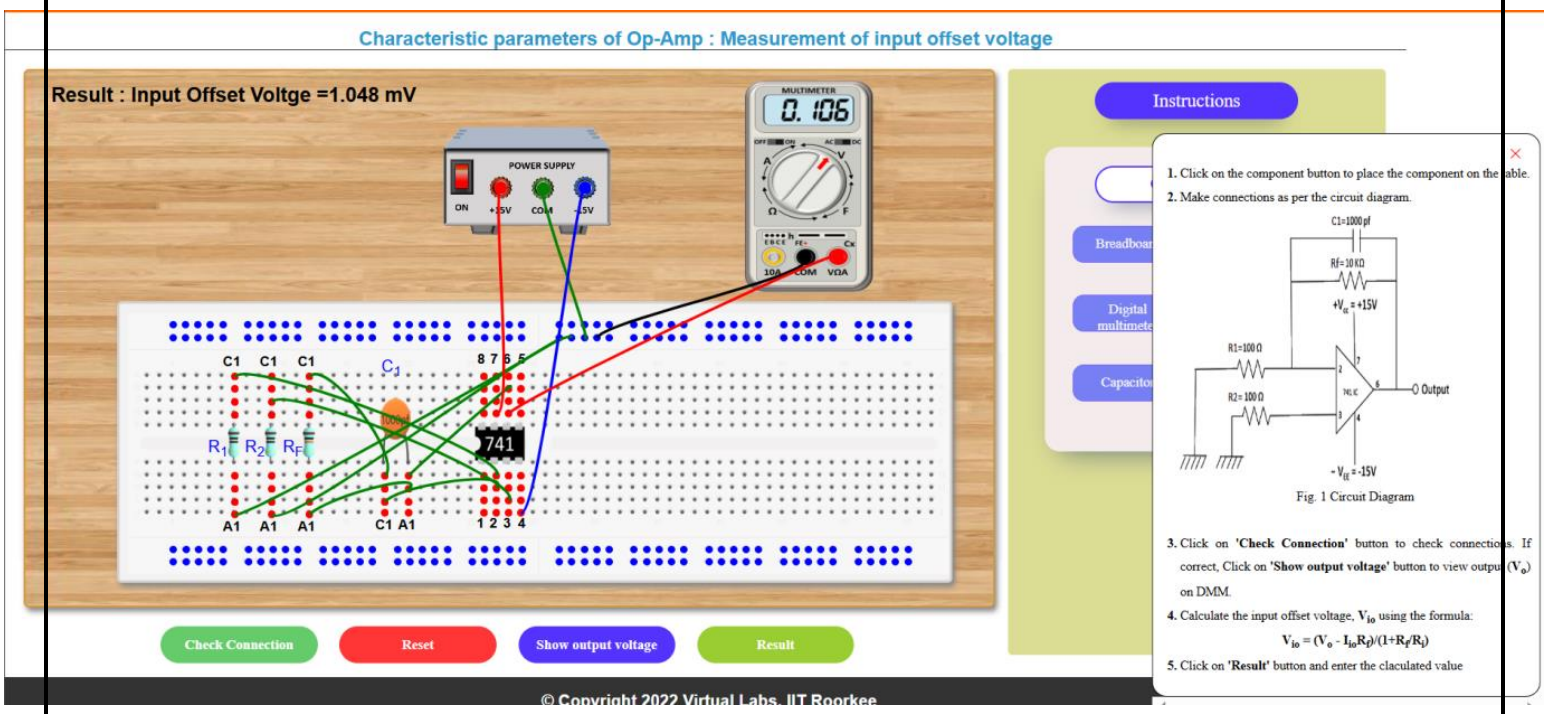
Vout=(1+RfR1)×VioV_{out} = (1 + \frac{R_f}{R_1}) \times V_{io}Vout =(1+R1Rf)×Vio

Hence, the input offset voltage is generally calculated as:

Vio=Vout1+RfR1V_{io} = \frac{V_{out}}{1 + \frac{R_f}{R_1}}Vio=1+R1 RfVout

Accordingly, by measuring the output voltage and knowing the closed-loop gain, the input offset voltage of the 741 op-amp can be determined.

Fig-3: Input offset voltage V_{io}



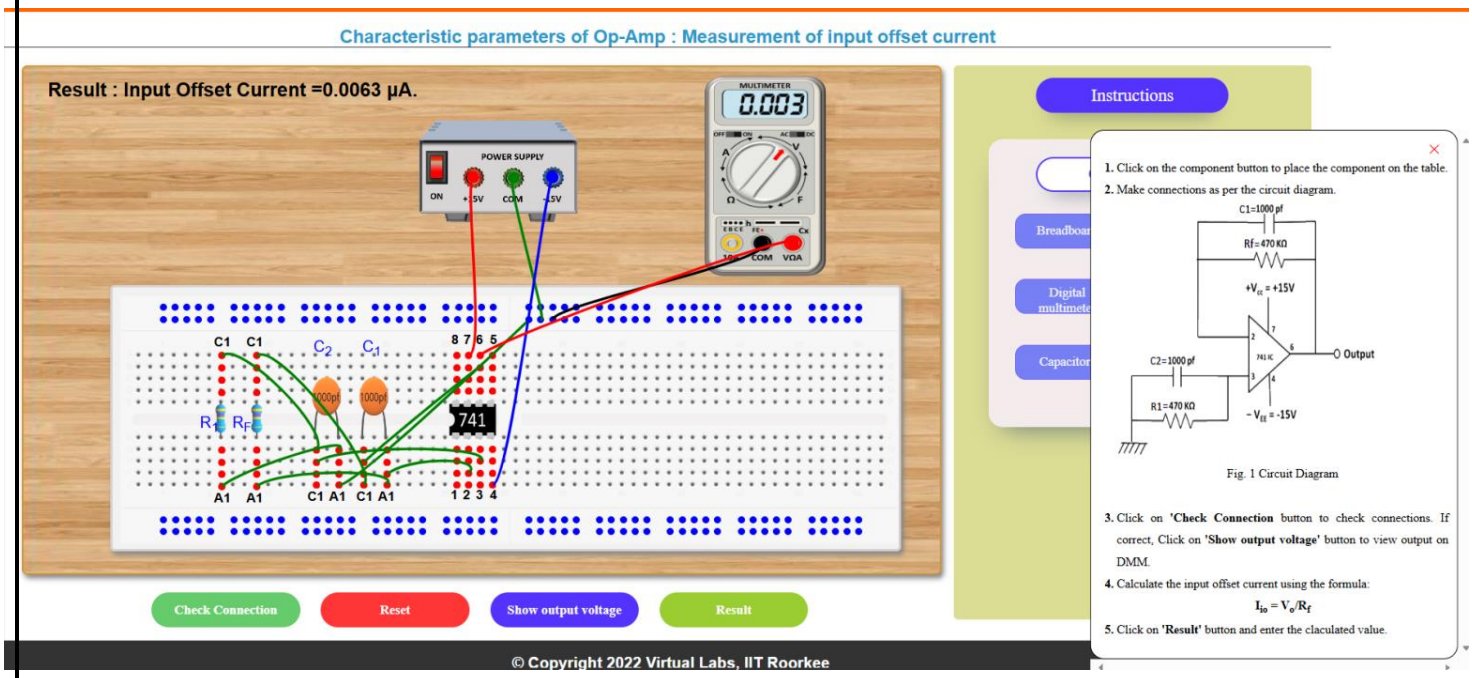
Input offset current

Measurement of Input Offset Current of 741 Op-Amp
To measure the input offset current (Iio) **of a 741 operational amplifier**

Measurement of Input Offset Current of 741 Op-Amp

- An ideal operational amplifier draws zero current at its input terminals. However, in a practical op-amp like the 741, small currents called input bias currents flow into the inverting and non-inverting terminals. Due to internal transis generallytor mis generallymatch, these two currents are not exactly equal. The difference between them is generally called the input offset current (Iio).
- In this generally experiment, the 741 is generally connected in a high-resis generallytance feedback configuration. Because the feedback resis generallytor value is generally large, even a very small offset current produces a measurable output voltage.
- The output voltage is generally related to the input offset current by:
- Vo=Iio×RfV_o = I_{io} \times R_fVo=Iio×Rf
- Hence, the input offset current is generally calculated as:
- Iio=VoRfI_{io} = \frac{V_o}{R_f}Iio=RfVo
- By measuring the output voltage and knowing the feedback resis generallytance, the input offset current of the 741 op-amp can be determined.

Fig-4: Input offset current I_{io}



Measurement of Inverting Bias Current of 741 Op-Amp

In an ideal op-amp, no current flows into the input terminals. But in a practical 741 op-amp, a small current called input bias current flows into both the inverting and non-inverting terminals due to internal transis generallytor operation.

In this generally experiment, the inverting bias current is generally measured by connecting a large feedback resis generallytor (Rf) between the output and the inverting input, while the non-inverting input is generally grounded. Because Rf has a high value, even a very small bias current produces a measurable output voltage.

The output voltage is generally related to the inverting bias current by:

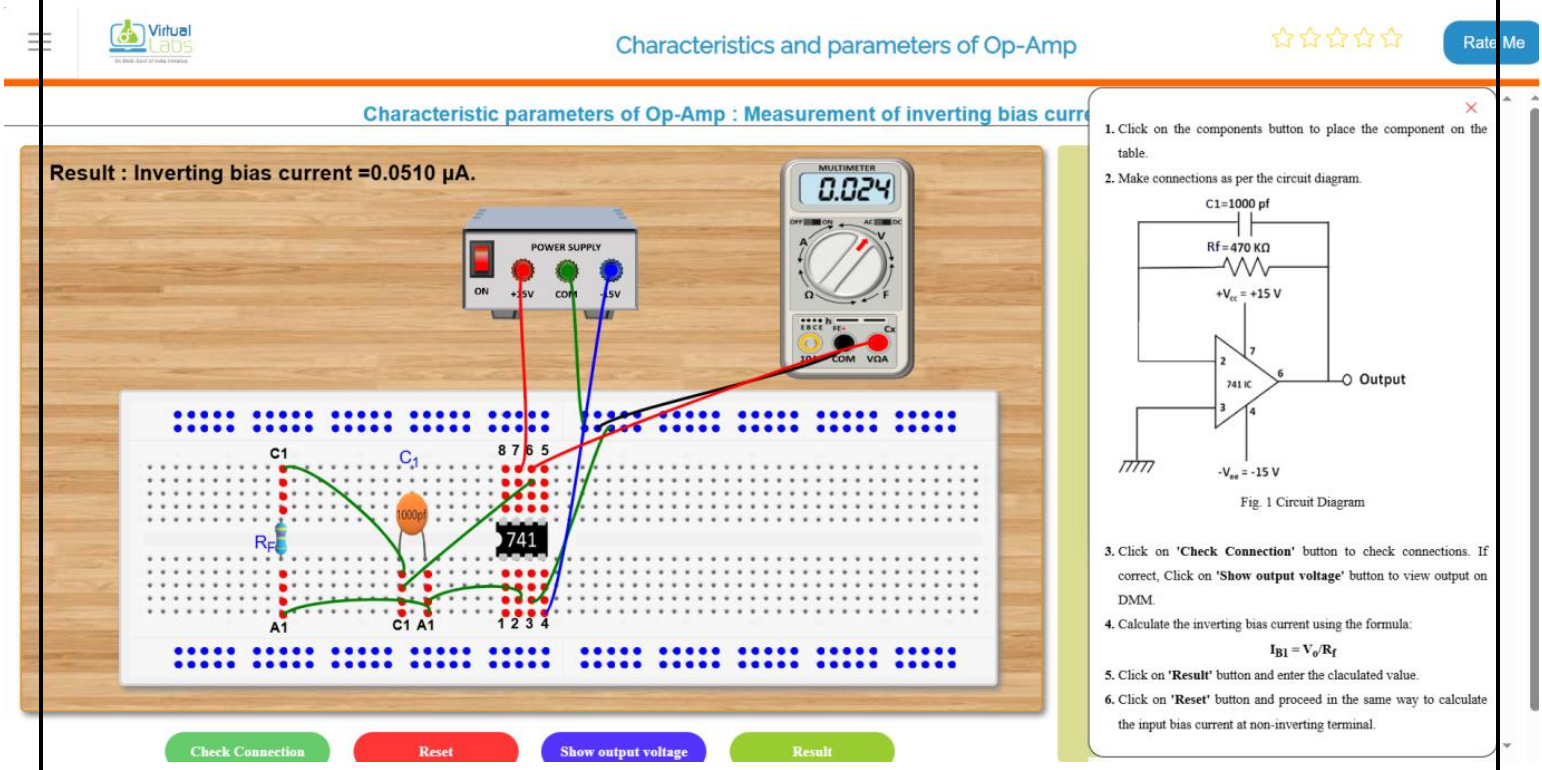
Vo=IB1×RfV_o = I_{B1} \times R_fVo=IB1×Rf

Hence, the inverting bias current is generally calculated as:

IB1=VoRfI_{B1} = \frac{V_o}{R_f}IB1=RfVo

By measuring the output voltage and knowing the feedback resis generallytance, the inverting bias current of the 741 op-amp can be determined.

Fig-4: Input bias current at inverting terminal



Measurement of Non-Inverting Bias Current of 741 Op-Amp

In a practical 741 op-amp, a small current flows into both input terminals. This generally current is generally called input bias current. The current flowing into the non-inverting terminal is generally called the non-inverting bias current (IB2).

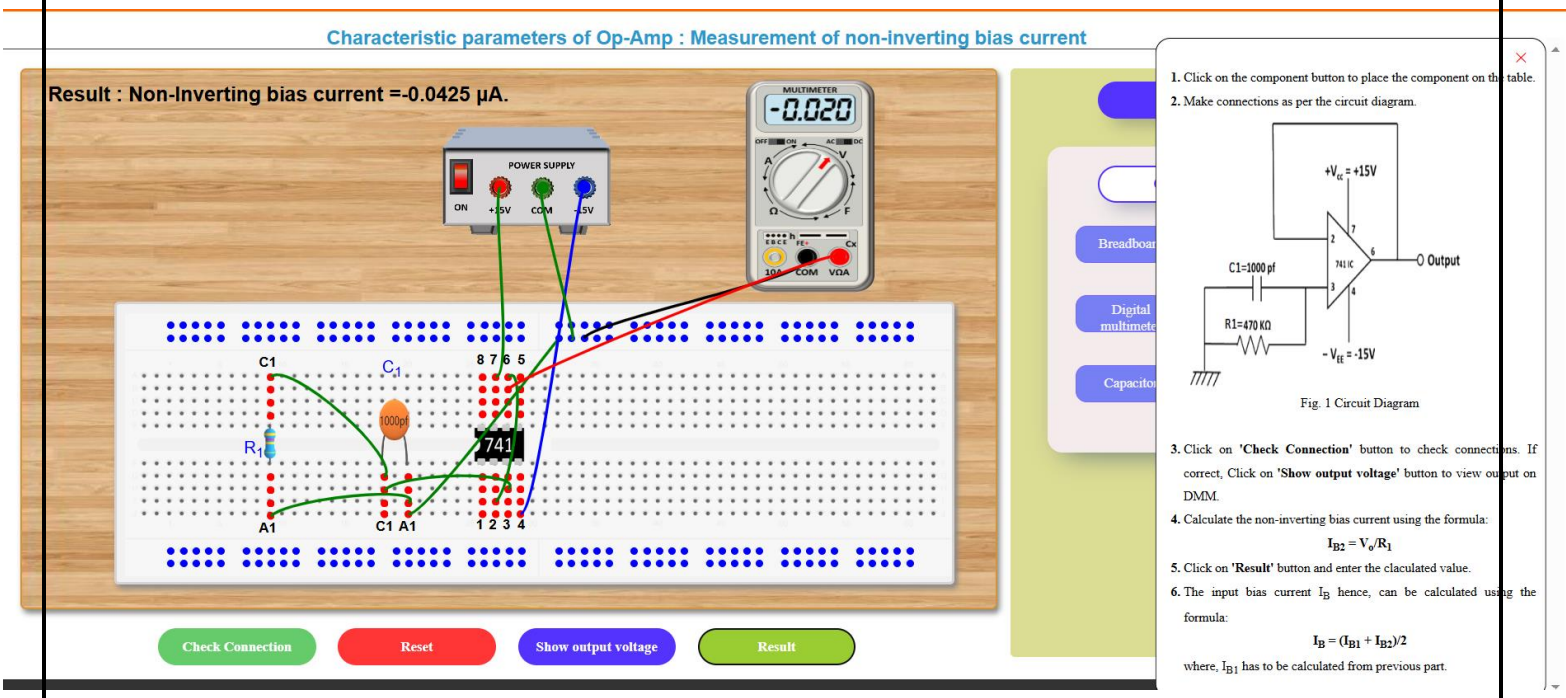
In this generally experiment, the inverting input is generally properly connected, and a resis generallytor (R1) is generally connected at the non-inverting terminal. Due to the bias current flowing into the non-inverting input, a small voltage is generally developed across the resis generallytor. This generally voltage appears at the output.

Since the resis generallytor value is generally known, the non-inverting bias current is generally calculated using:

IB2=VoR1I_{B2} = \frac{V_o}{R_1}IB2=R1Vo

Accordingly, by measuring the output voltage and knowing the resis generallytance value, the non-inverting bias current of the 741 op-amp can be determined.

Fig-4: Input bias current at Non inverting terminal



Short Theory – Measurement of Slew Rate of 741 Op-Amp

Slew rate is generally defined as the maximum rate of change of output voltage with respect to time. It indicates how fast an op-amp can respond to rapid changes in the input signal. It is generally expressed in volts per microsecond (V/µs).

In this generally experiment, a high-frequency sinusoidal signal is generally applied to the 741 op-amp. As the frequency increases, the output waveform starts to dis generallytort and becomes triangular. This generally happens because the op-amp cannot change its output voltage fast enough, and the maximum slope of the output waveform represents the slew rate.

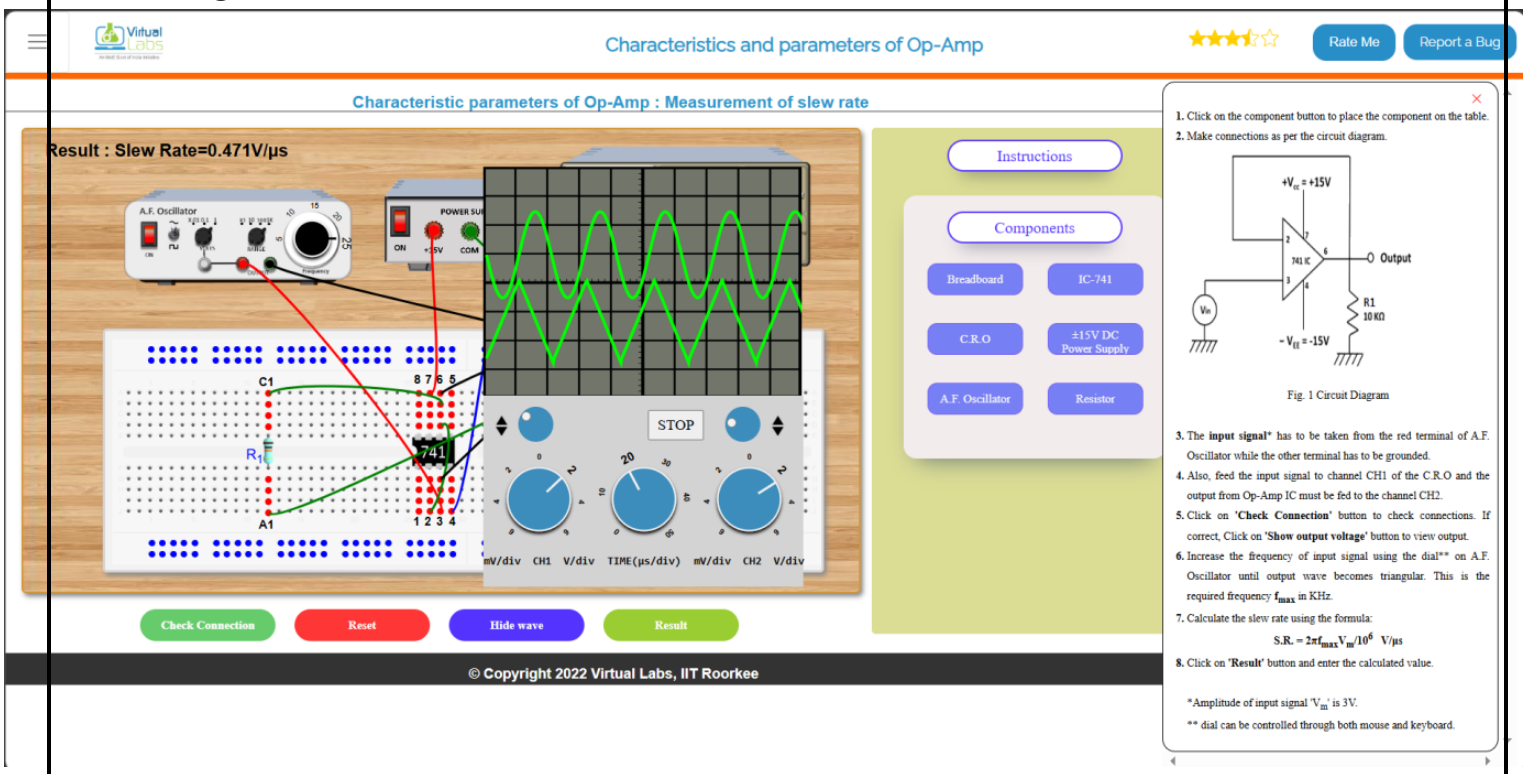
The slew rate is generally calculated using:

[
S.R. = 2\pi f_{max} V_m
]

where
(f_{max}) = frequency at which dis generallytortion begins,
(V_m) = peak amplitude of output signal.

By observing the waveform on the CRO and measuring the required values, the slew rate of the 741 op-amp can be determined.

Fig-4: Slew Rate



Important Points

- The IC 741 requires dual power supply for proper operation.
- The typical input offset voltage of 741 is in millivolts range.
- The input bias current is generally in nanoampere range.
- Slew rate of IC 741 is typically around 0.5 V/µs.
- The gain bandwidth product of 741 is approximately 1 MHz.
- IC 741 is internally frequency compensated.

Explanation of (Slew Rate Graph)

The last figure represents the slew rate measurement waveform observed on the CRO. When a high frequency input signal is applied, the output waveform begins to distort and appears triangular in shape. This distortion occurs because the op-amp cannot change its output voltage instantaneously. The maximum slope of this triangular waveform indicates the slew rate of the IC 741 operational amplifier.